**Quiz 4 - DSC 10, Spring 2026**

Full Name:

PID:

Quiz Time: ◯ 3PM ◯ 3:30PM ◯ 4PM ◯ 4:30PM

**Instructions:**

- This quiz consists of 3 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.
  ◯ A bubble means that you should only **select one choice**.
  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.
- Assume we have already run `import babypandas as bpd` and `import numpy as np`.

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By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

# Version A

Please do not open your quiz until instructed to do so.

Comic-Con is a massive pop culture and entertainment convention held each summer at the San Diego Convention Center. UC San Diego is capitalizing on its proximity to this major event and is offering Comic Con weekend accommodations in empty dorm rooms.

UCSD housing coordinators Pranav and Ray each processed a random sample of 25 Comic Con housing reservations. The DataFrame `res` has 50 rows, each of which represents a reservation processed by Pranav or Ray. Below are column descriptions and a preview of the first few rows of the DataFrame.

- `"coordinator"` (`str`): either `"Pranav"` or `"Ray"`
- `"package"` (`str`): either `"early_bird"` or `"standard"`
- `"dates"` (`str`): either `"Jul22-26"` or `"Jul23-27"`
- `"price"` (`float`): `760.50` for `"early_bird"` and `845.00` for `"standard"`

| | coordinator | package | dates | price |
|---|---|---|---|---|
| 0 | Ray | standard | Jul22-26 | 845.0 |
| 1 | Ray | early_bird | Jul22-26 | 760.5 |
| 2 | Pranav | early_bird | Jul23-27 | 760.5 |
| 3 | Pranav | early_bird | Jul22-26 | 760.5 |
| 4 | Ray | early_bird | Jul23-27 | 760.5 |

## Question 1

**a)** The number of reservations for each of the `"dates"` processed by each coordinator are given in the table to the right.

| | `"Pranav"` | `"Ray"` |
|---|---|---|
| `"Jul22-26"` | 20 | 18 |
| `"Jul23-27"` | 5 | 7 |

Compute the total variation distance (TVD) between Pranav's distribution of `"dates"` and Ray's distribution of `"dates"`. Give your answer as an **exact decimal or simplified fraction**.

$$0.08 = \frac{2}{25}$$

**b)** The table to the right shows how many reservations of each `"package"` type were processed by each coordinator.

| | `"Pranav"` | `"Ray"` |
|---|---|---|
| `"early_bird"` | `p` | `r` |
| `"standard"` | `25 - p` | `25 - r` |

Which of the following expressions correctly computes the TVD between Pranav's distribution of `"package"` type and Ray's distribution of `"package"` type?

- ◯ `(p - r) / 25`
- ● `abs(p - r) / 25`
- ◯ `0.5 * abs(p - r) / 25`
- ◯ `0.5 * sum(abs(p / 25 - r / 25))`
- ◯ `0.5 * (abs(p - r) + abs((25 - p) - (25 - r)))`

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## Question 2

Pranav is doing a hypothesis test with the following null hypothesis: in the population of reservations, each reservation has a 30% chance of being `"early_bird"`. His test statistic is the **percentage** of `"early_bird"` reservations, and his observed data is **all** of the data in `res`. Write one line of code to calculate one simulated value of the test statistic under the assumptions of the null. Your code should produce an **integer between 0 and 100**.

**Solution:** `np.random.multinomial(50, [0.3, 0.7])[0] / 50 * 100`

## Question 3

The housing director, Ella, wants to know whether Ray's reservations have significantly higher prices than Pranav's. Ella has access to `res` but no knowledge of the population distribution of reservation prices.

**a)** Which pair of hypotheses should be used for Ella's test? Select the best answer.

- ◯ Null: The mean price for Ray equals the mean price for Pranav.
  Alt: The mean price for Ray does not equal the mean price for Pranav.
- ◯ Null: Ray's prices come from a population with mean \$802.75.
  Alt: Ray's prices come from a population with mean larger than \$802.75.
- ● Null: Pranav and Ray's reservations come from the same price distribution.
  Alt: Ray's reservations come from a price distribution with a larger mean than Pranav's.

**b)** Which of the following approaches are appropriate for Ella's goal? Select all that apply.

- ■ Run a permutation test by shuffling coordinator labels and using the difference in mean prices as the test statistic.
- ☐ Run a permutation test by shuffling prices and using the absolute difference in mean prices as the test statistic.
- ☐ Bootstrap a 95% confidence interval for the difference in mean prices, then check whether 0 is in the interval.
- ■ Run a standard hypothesis test to see whether Ray's sample mean looks like the mean of a simple random sample taken from `res`, or whether it is too high.
- ☐ None of the above.

**c)** Suppose Ella performs a test where the test statistic is Pranav's mean price minus Ray's mean price. The observed value of the test statistic is $-50.7$. Which simulated statistics should be counted when computing the p-value?

- ◯ Simulated statistics greater than or equal to $-50.7$.
- ● Simulated statistics less than or equal to $-50.7$.
- ◯ Simulated statistics whose absolute value is greater than or equal to 50.7.
- ◯ Simulated statistics whose absolute value is less than or equal to 50.7.

**Reminders:**

- **Write your PID** on the front page and next to the PID line in the header on pages where it appears.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.